

Linear Regression

$$y = X\beta + E \quad \hat{\beta} = (X^T X)^{-1} X^T y$$

$$SS_{Res} = L = |y - X\hat{\beta}|^2 = y^T y - \hat{\beta}^T X^T y$$

$$= y^T y - n\bar{y}^2 - [\hat{\beta}^T X^T y - n\bar{y}^2]$$

$$\sum (y - \hat{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum (y - \hat{y})^2$$

$$SS_{Total} = SS_{Regression} + SS_{Res}$$

$$\frac{\partial L}{\partial \beta} = 0$$

Covariance Matrix

$$(X^T X)^{-1} \sigma^2 = \text{Var}(\hat{\beta})$$

$$= SE(\hat{\beta})^2$$

$$\hat{\sigma}^2 = \frac{SSE}{n-p}$$

unbiased estimator
n = no. of obs
p = df = no. of wts

Metrics (only we actual data used to fit)

$$R^2 = \frac{SS_{Reg}}{SS_{Total}}$$

df = p-1

$$R_{adj} = 1 - \left(\frac{SS_{Res}}{n-p} \right) / \left(\frac{SS_{Total}}{n-1} \right) \in [-1, 1]$$

High Better
worse than mean predictor

df [matrix] = no. of elements

X: n x p
X^TX: p x p
y: n x 1
β: p x 1

PRESS [leave-one-out cross validation]
Predicted Residual Error Sum of Squares

for OLS only
one fit is enough

$$H = X(X^T X)^{-1} X^T \text{ hat matrix } [y = H y]$$

$$e_i = y_i - \hat{y}_i$$

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}$$

Low Better

$$PRESS = \sum \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

∈ [0, ∞]

$$R^2_{PRESS} = 1 - \frac{PRESS}{SST}$$

MVA
A = LU: Solve Ax = b ⇒ Ly = b, Ux = y
[now reduced] Cholesky Decomp (symm pos definite)
LL^T = A, Idempotent: A = A² = A³ ...
L_{ij} = √(A_{ij} - ∑_{k=1}ⁱ⁻¹ L_{ik}L_{jk})
L_{ij} = 1/L_{ii} (A_{ij} - ∑_{k=1}ⁱ⁻¹ L_{ik}L_{jk})
One-hot Encoding (1-column omitted)
Dummy Var

$$z = \frac{\sigma}{\sigma} \text{ (population)}$$

$$R_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \rightarrow \text{Corr}(x, y)$$

$$\Sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix}$$

$$S = \frac{|x - \bar{x}|^2}{n-1} \text{ (sample)}$$

$$R = \frac{\tilde{X}^T \tilde{X}}{n-1} \text{ normalized}$$

Statistical Distance centre (0,0)

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \frac{X_1^2}{\sigma_{11}} + \frac{X_2^2}{\sigma_{22}} = d^2$$

Ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

$$c^2 = a^2 - b^2$$

[dist: centre → focus]

Mahalanobis and Chi-Squared

$$(x - \mu)^T \Sigma^{-1} (x - \mu) \leq \chi^2(\alpha)$$

with probability 1 - α

$$z_1^2 + z_2^2 + \dots + z_p^2 = \chi_p^2$$

Singular = no inverse

$$d_i^2 = \left(1 - \frac{1}{R_{ii}} \right) \left[\frac{(x_{i1} - \mu_1)^2}{\sigma_{11}} + \frac{(x_{i2} - \mu_2)^2}{\sigma_{22}} - \frac{2R_{12}(x_{i1} - \mu_1)(x_{i2} - \mu_2)}{\sigma_{11}\sigma_{22}} \right]$$

extra cov due to

Box Plot whiskers = 1.5 x IQR
Median Threshold: 3x Median Absolute Deviation